

136. Given below are two statements:

Statement I: The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II: Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

137. Match **List I** with **List II**.

List I	List II
A. Adenosine	I. Nitrogen base
B. Adenylic acid	II. Nucleotide
C. Adenine	III. Nucleoside
D. Alanine	IV. Amino acid

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-III, C-I, D-IV

138. Consider the following:

- A. The reductive division for the human female gametogenesis starts earlier than that of the male gametogenesis.
- B. The gap between the first meiotic division and the second meiotic division is much shorter for males compared to females.
- C. The first polar body is associated with the formation of the primary oocyte.
- D. Luteinizing Hormone (LH) surge leads to disintegration of the endometrium and onset of menstrual bleeding.

Choose the **correct** answer from the options given below:

- (1) A and B are true
- (2) A and C are true
- (3) B and D are true
- (4) B and C are true

139. All living members of the class Cyclostomata are:

- (1) Free living
- (2) Endoparasite
- (3) Symbiotic
- (4) Ectoparasite

140. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R): Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

141. Match **List I** with **List II**.

List I	List II
A. Scutellum	I. Persistent nucellus
B. Non-albuminous seed	II. Cotyledon of Monocot seed
C. Epiblast	III. Groundnut
D. Perisperm	IV. Rudimentary cotyledon

Choose the option with all **correct** matches.

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-IV, C-III, D-I

142. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): All vertebrates are chordates but all chordates are not vertebrate.

Reason (R): The members of subphylum vertebrata possess notochord, during the embryonic period, the notochord is replaced by a cartilaginous or bony vertebral column in adults.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) **A** is false but **R** is true

143. Identify the statement that is **NOT** correct.
- (1) Each antibody has two light and two heavy chains.
 - (2) The heavy and light chains are held together by disulfide bonds.
 - (3) Antigen binding site is located at C-terminal region of antibody molecules.
 - (4) Constant region of heavy and light chains are located at C-terminus of antibody molecules.

144. Silencing of specific mRNA is possible via RNAi because of -
- (1) Complementary dsRNA
 - (2) Inhibitory ssRNA
 - (3) Complementary tRNA
 - (4) Non-complementary ssRNA

145. Genes R and Y follow independent assortment. If RRY YY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F₂ generation?
- (1) Phenotypic ratio - 1 : 2 : 1
 - (2) Phenotypic ratio - 3 : 1
 - (3) Phenotypic ratio - 9 : 3 : 3 : 1
 - (4) Phenotypic ratio - 9 : 7

146. Histones are enriched with -
- (1) Lysine & Arginine
 - (2) Leucine & Lysine
 - (3) Phenylalanine & Leucine
 - (4) Phenylalanine & Arginine

147. The first menstruation is called:
- (1) Menopause
 - (2) Menarche
 - (3) Diapause
 - (4) Ovulation

148. Match List - I with List - II.

List - I	List - II
A. Heart	I. Erythropoietin
B. Kidney	II. Aldosterone
C. Gastro-intestinal tracts	III. Atrial natriuretic factor
D. Adrenal Cortex	IV. Secretin

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-III, C-IV, D-II
- (4) A-III, B-I, C-IV, D-II

149. The protein portion of an enzyme is called:

- (1) Cofactor
- (2) Coenzyme
- (3) Apoenzyme
- (4) Prosthetic group

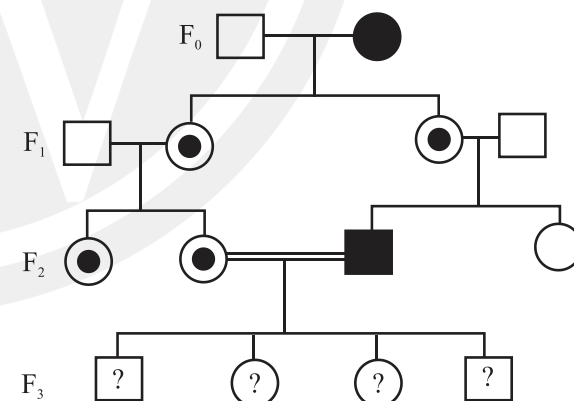
150. Which of the following is the unit of productivity of an Ecosystem?

- (1) gm⁻²
- (2) KCal m⁻²
- (3) KCal m⁻³
- (4) (KCal m⁻²)yr⁻¹

151. Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.

- (1) Analogy, convergent
- (2) Homology, divergent
- (3) Homology, convergent
- (4) Analogy, divergent

152. With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F₃ generation.



- Unaffected male
 ■ Affected male
 ○ Carrier female
 ○ Unaffected female
 ● Affected female

- (1) 1/4
- (2) 1/2
- (3) 1/8
- (4) Zero

153. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.
Reason (R): Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:
 (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
 (2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
 (3) **A** is true but **R** is false
 (4) **A** is false but **R** is true
154. How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?
 (1) 2 Meiosis and 3 Mitosis
 (2) 1 Meiosis and 2 Mitosis
 (3) 1 Meiosis and 3 Mitosis
 (4) No Meiosis and 2 Mitosis
155. Which of the following is an example of a zygomorphic flower?
 (1) Petunia
 (2) Datura
 (3) Pea
 (4) Chilli
156. After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s)/tissue(s) like:
 A. thymus
 B. bone marrow
 C. spleen
 D. lymph nodes
 E. Peyer's patches
 Choose the *correct* answer from the options given below:
 (1) B, C, D only
 (2) A, B, C only
 (3) E, A, B only
 (4) C, D, E only
157. Given below are two statements:
Statement I: Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.
Statement II: Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.
 In the light of the above statements, choose the **most appropriate** answer from the options given below:
 (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct
158. What is the main function of the spindle fibers during mitosis?
 (1) To separate the chromosomes
 (2) To synthesize new DNA
 (3) To repair damaged DNA
 (4) To regulate cell growth
159. Which one of the following is the characteristic feature of gymnosperms?
 (1) Seeds are enclosed in fruits.
 (2) Seeds are naked.
 (3) Seeds are absent.
 (4) Gymnosperms have flowers for reproduction.
160. Consider the following statements regarding function of adrenal medullary hormones:
 A. It causes pupillary constriction
 B. It is a hyperglycemic hormone
 C. It causes piloerection
 D. It increases/strength of heart contraction
 Choose the **correct** answer from the options given below:
 (1) C and D Only
 (2) B, C and D Only
 (3) A, C and D Only
 (4) D Only
161. Why can't insulin be given orally to diabetic patients?
 (1) Human body will elicit strong immune response
 (2) It will be digested in Gastro-Intestinal (GI) tract
 (3) Because of structural variation
 (4) Its bioavailability will be increased

162. Match List-I with List-II.

List-I		List-II	
A.	Pteridophyte	I.	<i>Salvia</i>
B.	Bryophyte	II.	<i>Ginkgo</i>
C.	Angiosperm	III.	<i>Polytrichum</i>
D.	Gymnosperm	IV.	<i>Salvinia</i>

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

163. Who proposed that the genetic code for amino acids should be made up of three nucleotides?

- (1) George Gamow
- (2) Francis Crick
- (3) Jacques Monod
- (4) Franklin Stahl

164. Match List-I with List-II.

List-I		List-II	
A.	The Evil Quartet	I.	Cryopreservation
B.	Ex situ conservation	II.	Alien species invasion
C.	<i>Lantana camara</i>	III.	Causes of biodiversity losses
D.	Dodo	IV.	Extinction

Choose the option with all **correct** matches.

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-IV, D-I

165. Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?

- (1) Luteinizing hormone (LH)
- (2) Anti-diuretic hormone (ADH)
- (3) Follicle-stimulating hormone (FSH)
- (4) Adenocorticotrophic hormone (ACTH)

166. Role of the water vascular system in Echinoderms is:

- A. Respiration and Locomotion
- B. Excretion and Locomotion
- C. Capture and transport of food
- D. Digestion and Respiration
- E. Digestion and Excretion

Choose the **correct** answer from the options given below:

- (1) A and B Only
- (2) A and C Only
- (3) B and C Only
- (4) B, D and E Only

167. Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body?

- (1) Acquired Immunity
- (2) Innate Immunity
- (3) Cell-mediated Immunity
- (4) Humoral Immunity

168. In bryophytes, the gemmae help in which one of the following?

- (1) Sexual reproduction
- (2) Asexual reproduction
- (3) Nutrient absorption
- (4) Gaseous exchange

169. In frog, the Renal portal system is a special venous connection that acts to link :

- (1) Liver and intestine
- (2) Liver and kidney
- (3) Kidney and intestine
- (4) Kidney and lower part of body

170. Given below are two statements:

Statement I: In ecosystem, there is unidirectional flow of energy of sun from producers to consumers.

Statement II: Ecosystems are exempted from 2nd law of thermodynamics.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

171. Which of the following statements about RuBisCO is true?
- It is active only in the dark.
 - It has higher affinity for oxygen than carbon dioxide.
 - It is an enzyme involved in the photolysis of water.
 - It catalyzes the carboxylation of RuBP.
172. Which of the following enzyme(s) are **NOT** essential for gene cloning?
- Restriction enzymes
 - DNA ligase
 - DNA mutase
 - DNA recombinase
 - DNA polymerase
- Choose the **correct** answer from the options given below :
- C and D only
 - A and B only
 - D and E only
 - B and C only
173. Read the following statements on plant growth and development.
- Parthenocarpy can be induced by auxins.
 - Plant growth regulators can be involved in promotion as well as inhibition of growth.
 - Dedifferentiation is a pre-requisite for re-differentiation.
 - Abscisic acid is a plant growth promoter.
 - Apical dominance promotes the growth of lateral buds.
- Choose the option with all correct statements.
- A, B, C only
 - A, C, E only
 - A, D, E only
 - B, D, E only
174. Which factor is important for termination of transcription?
- α (alpha)
 - σ (sigma)
 - ρ (rho)
 - γ (gamma)

175. Frogs respire in water by skin and buccal cavity and on land by skin, buccal cavity and lungs. Choose the **correct** answer from the following :
- The statement is true for water but false for land
 - The statement is true for both the environment
 - The statement is false for water but true for land
 - The statement is false for both the environment
176. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?
- They are monozygotic twins.
 - They are fraternal twins.
 - They were conceived through in vitro fertilization.
 - They have 75% identical genetic content.
177. Which of the following microbes is **NOT** involved in the preparation of household products?
- Aspergillus niger*
 - Lactobacillus*
 - Trichoderma polysporum*
 - Saccharomyces cerevisiae*
 - Propionibacterium sharmanii*
- Choose the **correct** answer from the options given below:
- A and B only
 - A and C only
 - C and D only
 - C and E only
178. Match **List-I** with **List-II**.
- | List-I | List-II |
|-----------------------------------|----------------------|
| A. Progesterone | I. Pars intermedia |
| B. Relaxin | II. Ovary |
| C. Melanocyte stimulating hormone | III. Adrenal Medulla |
| D. Catecholamines | IV. Corpus luteum |
- Choose the **correct** answer from the options given below:
- A-IV, B-II, C-I, D-III
 - A-IV, B-II, C-III, D-I
 - A-II, B-IV, C-I, D-III
 - A-III, B-II, C-IV, D-I

179. The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:

Statement I: The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II: The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

180. Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

(1) $\frac{dN}{dt} = r \left(\frac{K - N}{K} \right)$

(2) $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$

(3) $\frac{dN}{dt} = rN \left(\frac{N - K}{N} \right)$

(4) $\frac{dN}{dt} = N \left(\frac{r - K}{K} \right)$

